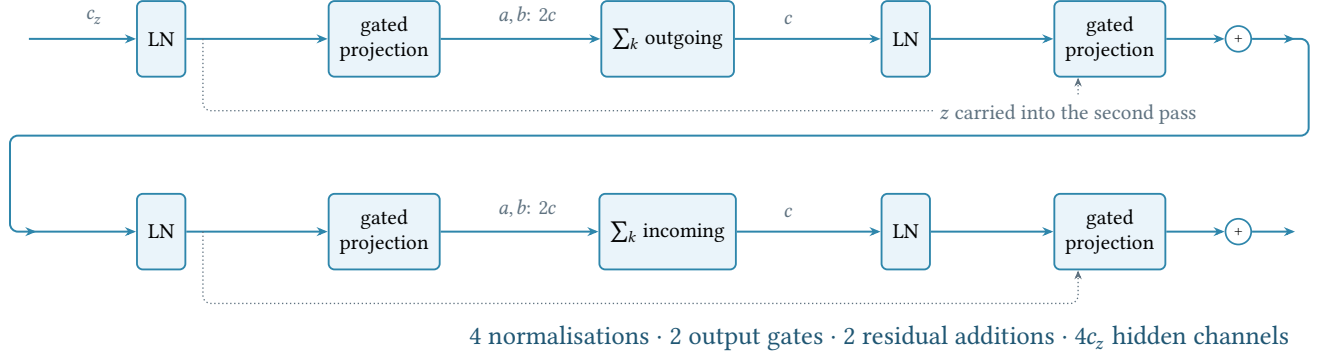


a AlphaFold 3: the same body executed twice



b MiniWorld 0: one body, forked only at the contraction

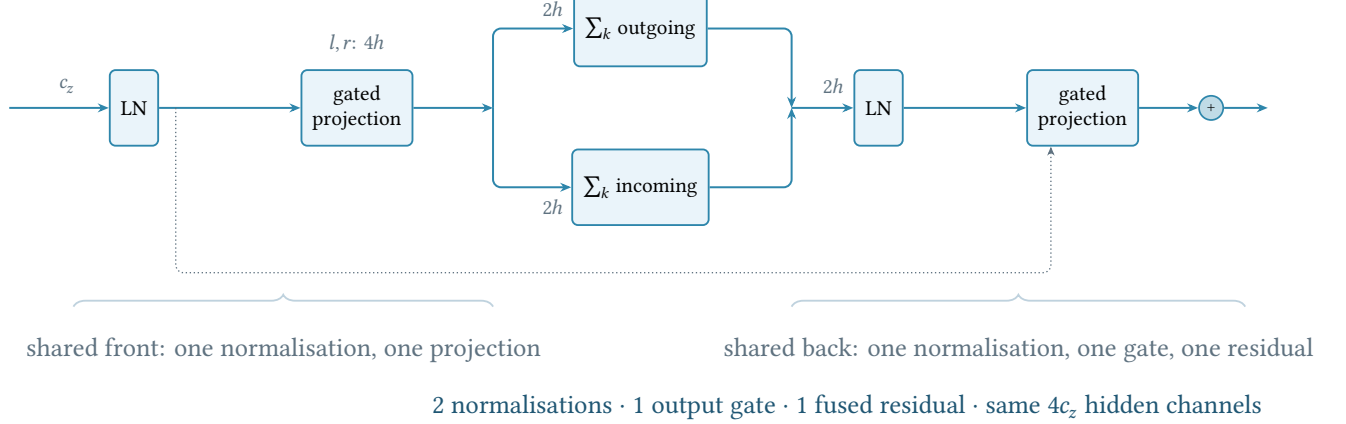


Figure S2. Dataflow inside the pair update. AlphaFold 3 (a) runs one body twice, threading z from the outgoing pass into the incoming pass, so every normalisation, gate, projection and residual is performed twice. MiniWorld 0 (b) runs the body once and forks only at the contraction — the single place where the two directions differ — so the normalisation and gated projection at the front and the normalisation, gate, projection and residual at the back are shared. Hidden capacity is unchanged: AlphaFold 3 forms $2c$ channels twice, MiniWorld 0 forms $4h$ once and splits it, and with $c = h = c_z = 128$ both amount to $4c_z$. The dotted line marks that the output gate reads the normalised input rather than the contraction, which is what lets one fused front feed both contractions.